

Phase 3.2 Metric Verification Report

Date: 2026-07-16 **Candidate Model:** physics_guided

1. Metric Audit Checklist

We cross-checked all values in the PDF reports against the raw data in `per_seed_metrics.csv` and `aggregate_metrics.csv`:

- **Mean RMSE:**

- Expected (per_seed_metrics): **1.60912** - Observed (Official Benchmark Report): **1.60912** - Status: **Match**

- **Mean MAE:**

- Expected: **1.27090** - Observed: **1.27090** - Status: **Match**

- **Paired t-test vs v1.3:**

- Expected: $t = 2.24336, p = 0.08829$

- Observed (Statistical Report): $t = 2.24336, p = 0.08829$

- Status: **Match**

- **Wilcoxon Signed-Rank p-value:**

- Expected: **0.06250** - Observed: **0.06250** - Status: **Match**

- **Cohen's d :**

- Expected: **0.72280** - Observed: **0.72280** - Status: **Match**

- **Bootstrap 95% Confidence Interval:**

- Expected: **[-0.01245, 0.12563]** - Observed: **[-0.01245, 0.12563]** - Status: **Match**

2. Discrepancy Analysis

No discrepancies exist in the regression or statistical metrics. Every reported metric matches the raw execution logs.